week1_notebook

September 22, 2026

1 HS4002 Week 1 Demo

1.1 Basics of Python

We're working within the **pandas** + **plotnine** framework.

We'll explore a toy dataset about *Love Is Blind* contestants to answer: **to what extent does age, gender, and race affect the likelihood of marriage in LIB?**

[25]:
```
import pandas as pd # import the packages we need
import numpy as np
import plotnine as p9
import warnings
warnings.filterwarnings('ignore')
```

1.2 Loading the Dataset

[26]:
```
file_location = "lib_toy.csv"
df = pd.read_csv(file_location)
```

2 What does a dataset usually look like?

[27]:
```
df.head() # give first 5 rows of df
```

[27]:

	Name	Season	Partner	Age	Married	Race	Gender	Unnamed: 7
0	Lauren Speed	1	Cameron Hamilton	32	1	Black	1	LIB + Lauren Speed
1	Cameron Hamilton	1	Lauren Speed	28	1	White	0	LIB + Cameron Hamilton
2	Amber Pike	1	Matthew Barnett	27	1	White	1	LIB + Amber Pike
3	Matthew Barnett	1	Amber Pike	27	1	White	0	LIB + Matthew Barnett
4	Giannina Gibelli	1	Damian Powers	25	0	Hispanic	1	LIB + Giannina Gibelli

3 Univariate Analysis

[28]:
```
df.describe() # generate descriptive statistics for variables
```

[28]:
```
            Season         Age     Married      Gender
count  120.000000  120.000000  120.000000  120.000000
mean     5.650000   30.708333    0.266667    0.500000
std      2.972281    3.090772    0.444071    0.502096
min      1.000000   24.000000    0.000000    0.000000
25%      3.000000   28.000000    0.000000    0.000000
50%      6.000000   31.000000    0.000000    0.500000
75%      8.000000   33.000000    1.000000    1.000000
max     10.000000   39.000000    1.000000    1.000000
```

[29]:
```
df['Race'].value_counts() # count the number of people for each race
```

[29]:
```
Race
White       67
Black       33
Hispanic    12
Asian        6
Mixed        2
Name: count, dtype: int64
```

3.1 How many seasons of LIB are in this dataset?

[30]:
```
df['Season'].value_counts()
```

[30]:
```
Season
10    16
4     14
7     14
1     12
2     12
9     12
3     10
5     10
6     10
8     10
Name: count, dtype: int64
```

3.2 How many contestants got married?

[31]:
```
df['Married'].value_counts()
```

[31]:
```
Married
0    88
```

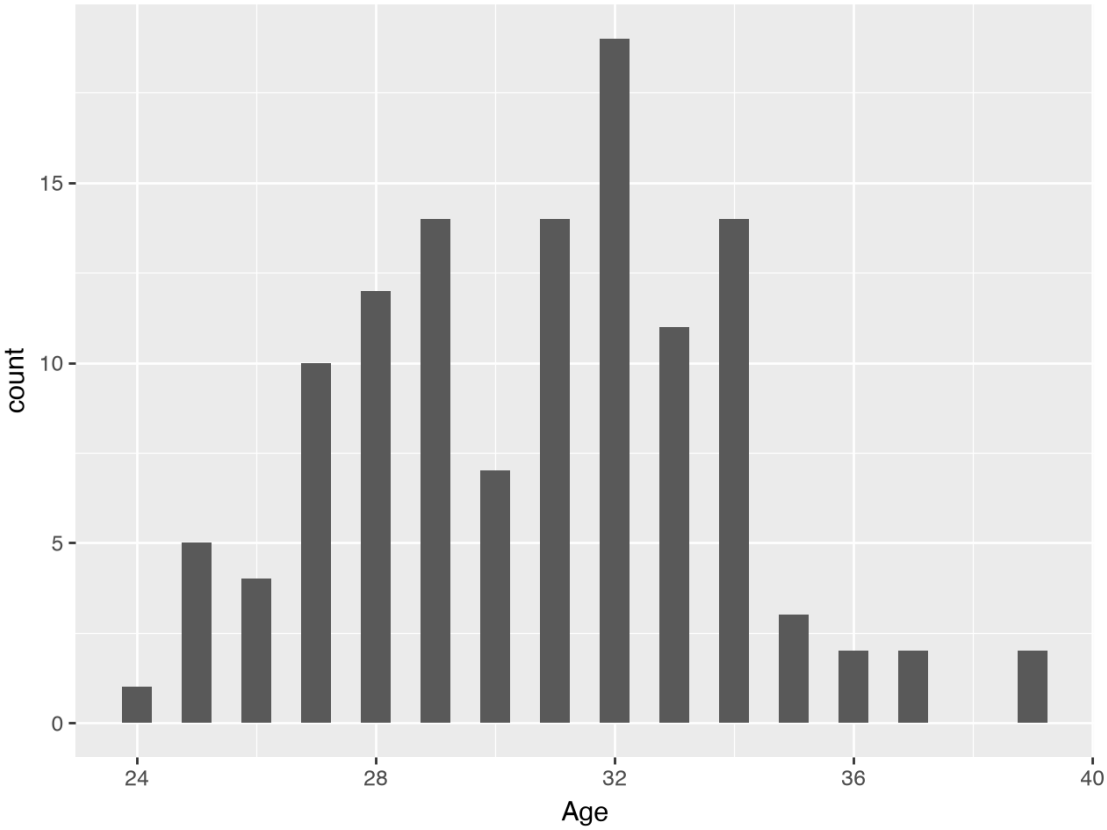
```
1    32
Name: count, dtype: int64
```

3.3 Univariate Plots

Let's make a histogram of the age distribution using `plotnine` — the grammar is identical to ggplot2. Note that column names in `aes()` are quoted strings in Python.

[32]:
```
(
    p9.ggplot(df) + # plot a graph using df
    p9.geom_histogram(p9.aes(x='Age'), binwidth=0.5) # plot a histogram of the␣
 ↪age of contestants
)
```

[32]:



[33]:
```
fig_fn = "ntu4eva.png"

p = (
    p9.ggplot(df) +
    p9.geom_histogram(p9.aes(x="Age"), binwidth=0.5, fill="red") +
    p9.theme_light() +
    p9.xlim(18, 40) + # set the limits for x-axis
```

```
    p9.labs(y="Count", x="Age", title="Age Distribution of Love is Blind␣
  ↪Contestants")
)
p.save(fig_fn, width=30, height=17, units="cm") # save this graph as an image␣
  ↪file!
p
```

[33]:



[34]:
```
fig_fn = "ntu4eva.png"

p = (
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age"), fill="steelblue") + # plot a density cruve␣
  ↪of age of contestants
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age", title="Age of Love is Blind Contestants")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

[34]:



4 Does Age Matter?

How does age relate to the likelihood of a contestant getting married?

[35]:
```
df.groupby('Married')['Age'].mean() # groups people by whether they are married␣
↪or not, then calculates the average age of each group
```

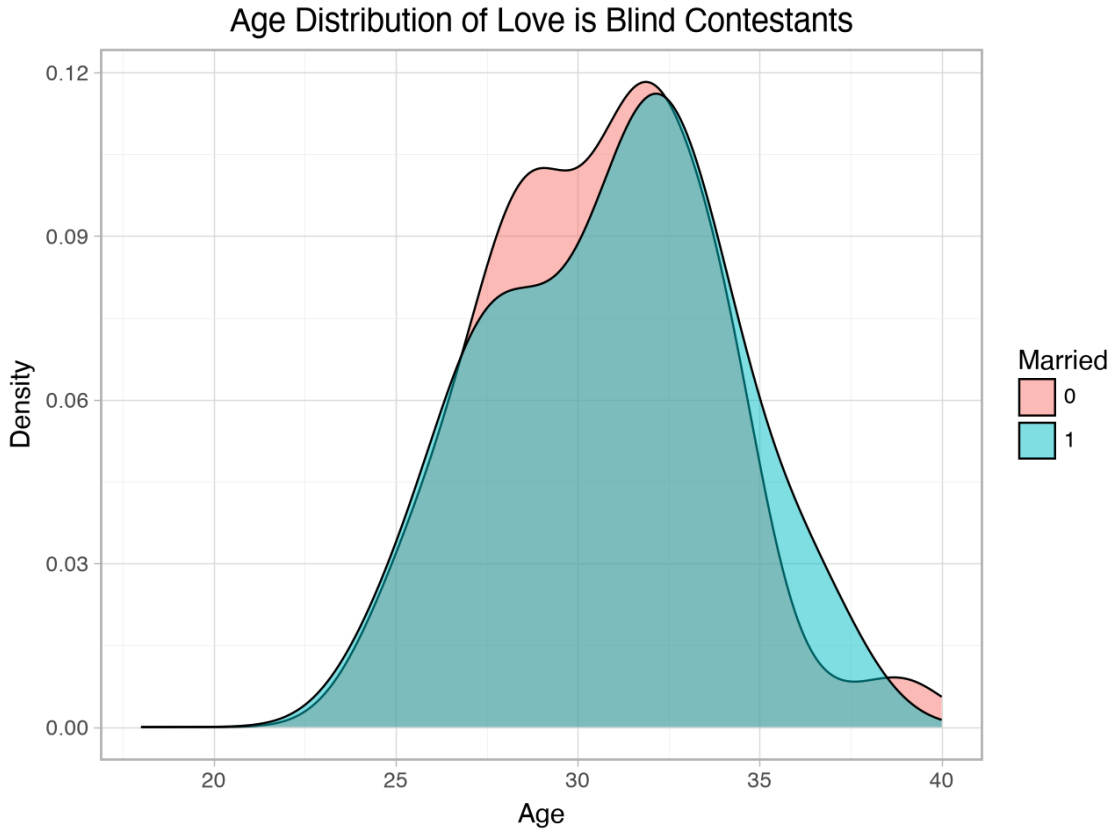
[35]:
```
Married
0    30.636364
1    30.906250
Name: Age, dtype: float64
```

[36]:
```
fig_fn = "age_by_maritalstatus.png"

p = (
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age", fill="factor(Married)"),␣
↪position="identity", alpha=0.5) + # postition = identity tells python to put␣
↪the graphs on top of each other, thus allowing overlapping
    p9.theme_light() +
```

```
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age",
         title="Age Distribution of Love is Blind Contestants",
         fill="Married") # name of the legend
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

[36]:



5 Let's Throw Race Into the Mix

[37]:
```
df.groupby(['Married', 'Race'])['Age'].mean()
```

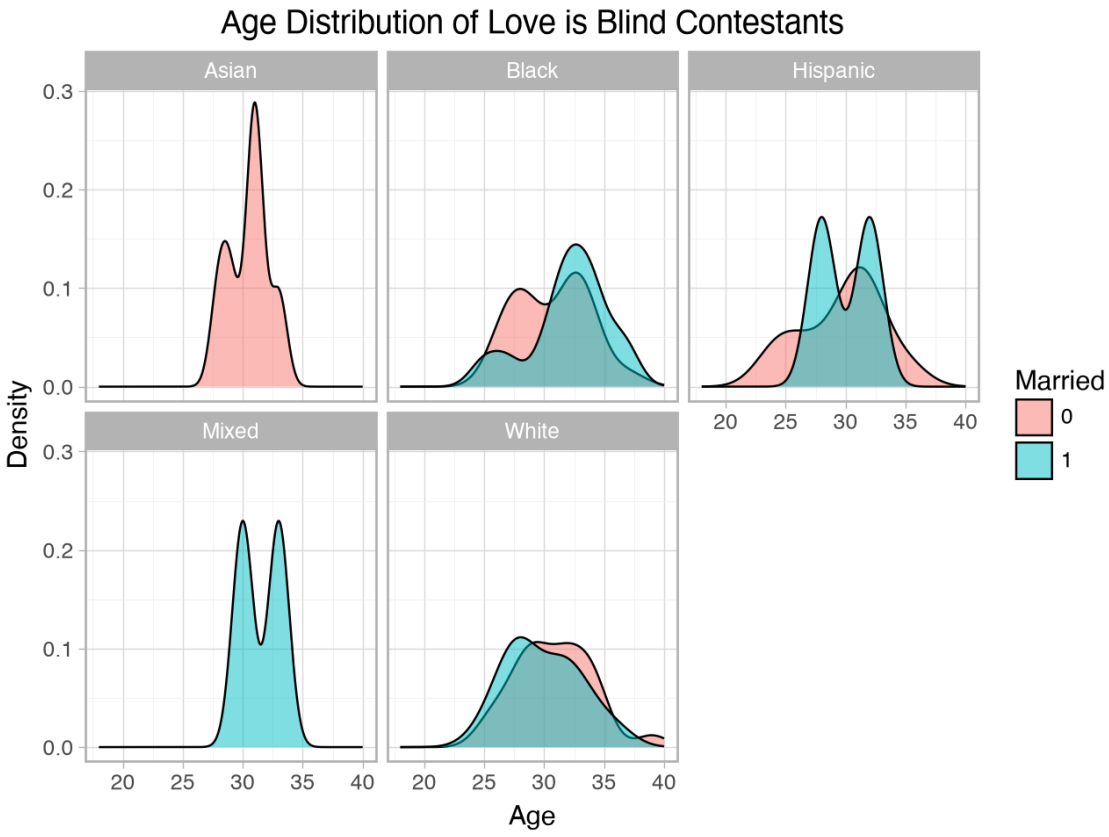
[37]:
```
Married  Race
0        Asian       30.500000
         Black       30.700000
         Hispanic    29.700000
         White       30.807692
1        Black       32.076923
         Hispanic    30.000000
         Mixed       31.500000
```

```
White       29.933333
Name: Age, dtype: float64
```

[38]:
```
fig_fn = "age_marital_race.png"

p = (
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age", fill="factor(Married)"),␣
 ↪position="identity", alpha=0.5) +
    p9.facet_wrap("Race") + # make a separate graph for each race!
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age",
          title="Age Distribution of Love is Blind Contestants",
          fill="Married")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

[38]:



6 Trying Adding Gender!

[39]:
```
df = df.assign(Gender=np.where(df['Gender'] == 1, 'Woman', 'Man'))
# rewrite the gender column. if it is 1, write as women, if not, write as man

df.groupby(['Married', 'Gender'])['Age'].mean()
```

[39]:
```
Married  Gender
0        Man       30.204545
         Woman     31.068182
1        Man       31.250000
         Woman     30.562500
Name: Age, dtype: float64
```

[40]:
```
fig_fn = "age_marital_gender.png"

p = (
    p9.ggplot(df) +
    p9.geom_density(p9.aes(x="Age", fill="factor(Married)"),␣
 ↪position="identity", alpha=0.5) +
    p9.facet_wrap("Gender") +
    p9.theme_light() +
    p9.xlim(18, 40) +
    p9.labs(y="Density", x="Age",
         title="Age Distribution of Love is Blind Contestants",
         fill="Married")
)
p.save(fig_fn, width=30, height=17, units="cm")
p
```

[40]:



[42]:
```
!jupyter nbconvert --to pdf week1_notebook.ipynb
```

```
[NbConvertApp] Converting notebook week1_notebook.ipynb to pdf
[NbConvertApp] Writing 35703 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 43444 bytes to week1_notebook.pdf
```